PROJECT SYNOPSIS

Employee Salary Prediction

Problem ID: 03

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Section: G-2

Course: Artificial Intelligence / Machine Learning
Academic Year: 2025 – 2029

Submitted To: Mr. Darshan Singh

Department of: B.Tech (cse ai and ml)
University name: Graphic era deemed to be university

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Employee Salary Prediction

Problem ID: 03

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Course: Artificial Intelligence / Machine Learning
Academic Year: 2025 – 2029

Submitted To: Mr. Darshan Singh

Department of: B.Tech (cse ai and ml)
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Employee Salary Prediction Using Regression-Based Machine Learning Models

An end-to-end machine learning pipeline applied to synthetic organizational data — bridging HR analytics and predictive modelling.

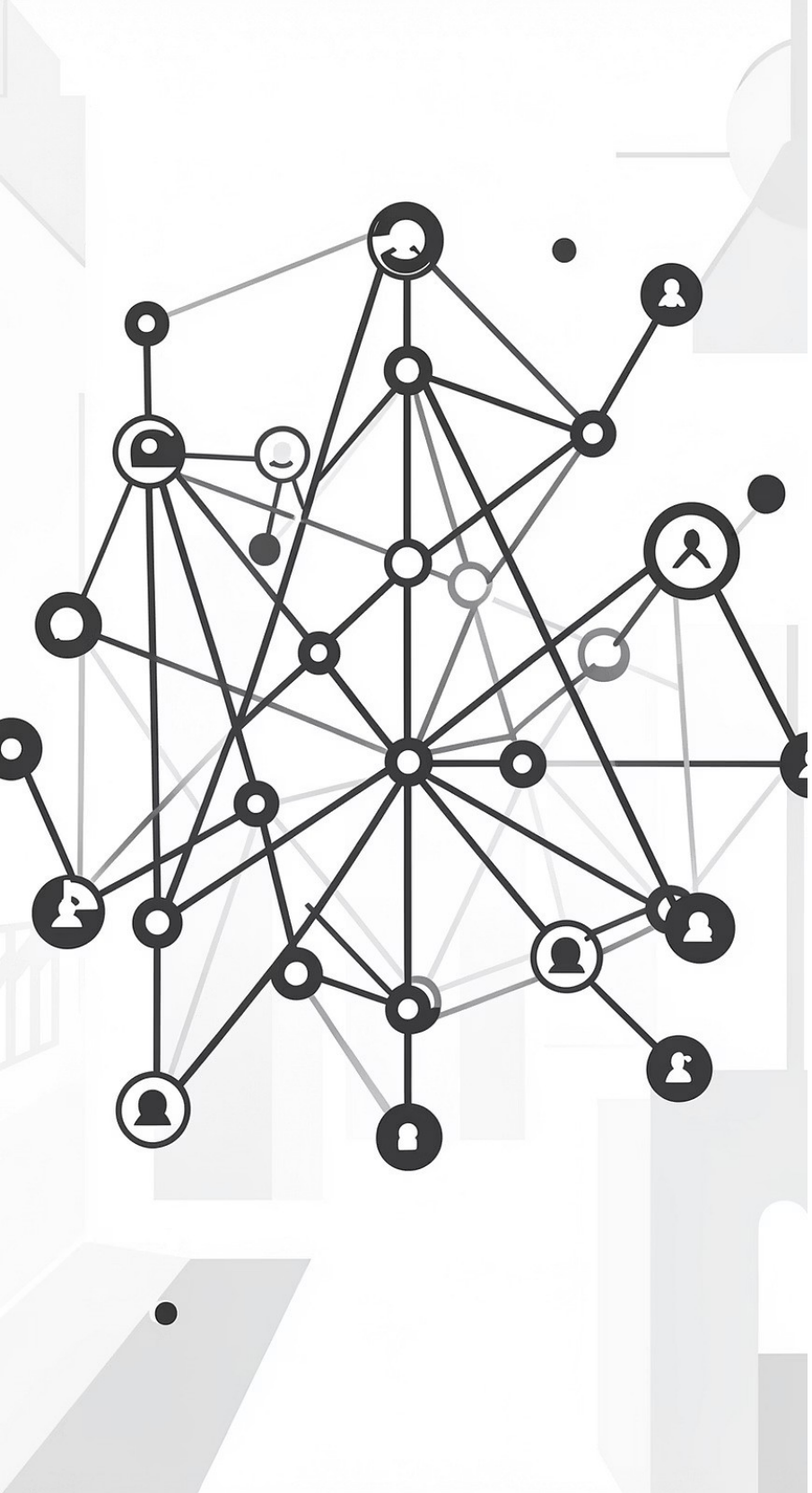
SYNOPSIS REPORT

Introduction

Accurate salary prediction is a critical capability for modern organisations. Misaligned compensation leads to talent attrition, pay inequity, and inflated labour costs — all of which erode organisational performance.

Machine learning offers a data-driven alternative to ad hoc salary-setting. By identifying the statistical relationships between employee attributes and compensation, regression models can provide transparent, reproducible, and scalable salary benchmarks.

> This project applies supervised regression techniques to a richly featured synthetic dataset, demonstrating a full ML pipeline applicable to real-world HR analytics contexts.



Problem Statement

Employee compensation is influenced by a multitude of interacting factors — experience, education, role, performance, and geography, among others. Traditional methods such as salary bands and manual benchmarking are often opaque, inconsistent, and unable to capture non-linear interactions between variables.

Complexity

Salary determinants span demographic, experiential, and organisational dimensions simultaneously.

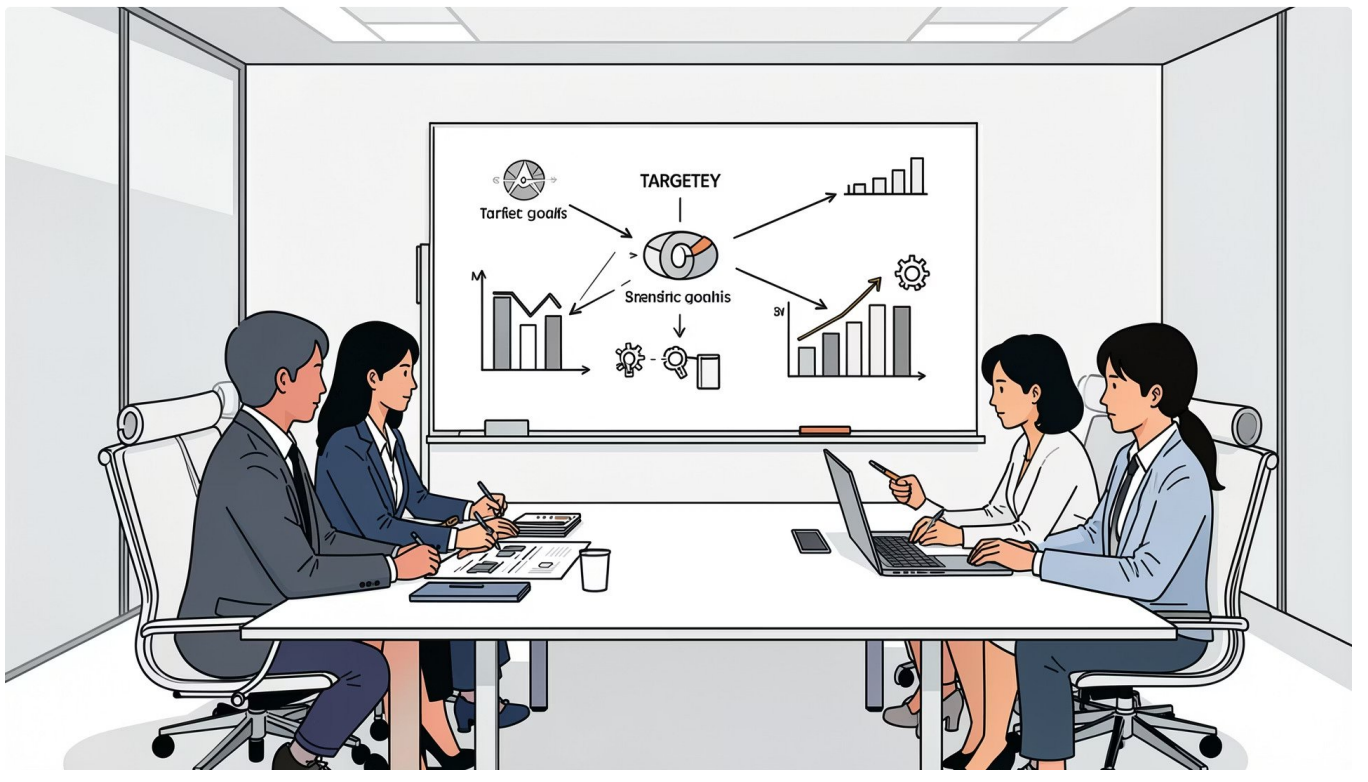
Inconsistency

Manual processes introduce bias and lack reproducibility across departments or HR practitioners.

Scalability

Organisations require automated, auditable systems capable of handling large, evolving workforces.

Project Objectives



01

Generate Synthetic Data

Construct a realistic 20-feature organisational dataset with a continuous salary target variable.

02

Perform EDA

Identify distributions, correlations, and outliers to inform preprocessing decisions.

03

Apply Feature Selection

Use Forward Selection to identify the most predictive subset of employee attributes.

04

Develop & Evaluate Models

Train and compare Linear, Ridge, and Lasso regression models using R^2 and RMSE metrics.

Dataset Description

A synthetic dataset of **1,000+ employee records** was programmatically generated to reflect realistic organisational structures. It contains **20 input features** and one continuous target variable: **Annual Salary (USD)**.

Demographic & Personal

- Age
- Gender
- Education Level
- Marital Status
- Number of Dependants

Experience & Performance

- Years of Experience
- Tenure at Current Company
- Number of Promotions
- Performance Score
- Training Hours Completed

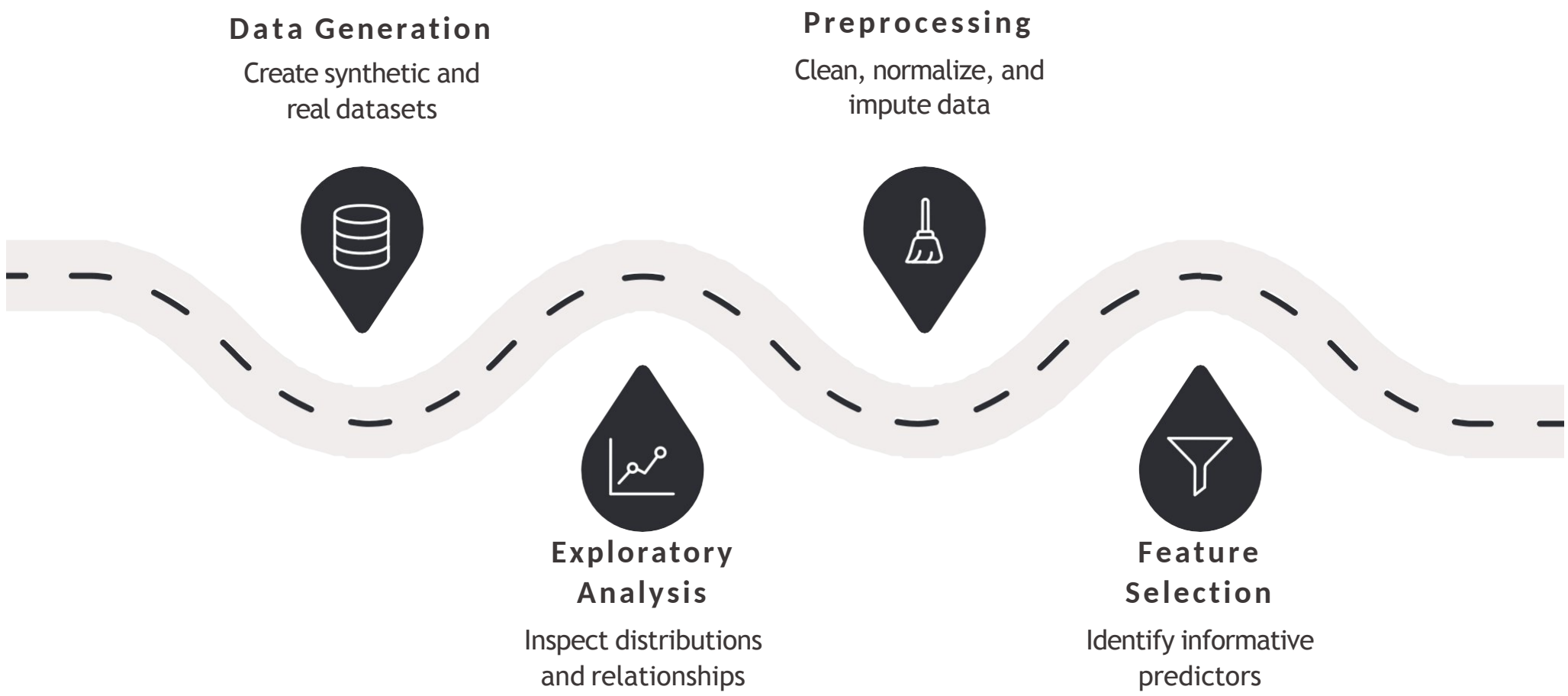
Organisational

- Department
- Job Title / Role
- Employment Type
- Location / Region
- Company Size

Compensation Indicators

- Previous Salary
- Bonus Percentage
- Overtime Hours
- Remote Work Flag
- Certifications Count

Methodology



The pipeline follows a structured, reproducible workflow — from synthetic data generation through to model evaluation — ensuring each stage informs the next with rigorous, data-driven decisions.

Methodology — Deep Dive

Synthetic Data Generation

Statistical distributions (normal, categorical, Bernoulli) model realistic employee attributes using NumPy and Faker libraries.

EDA

Correlation heatmaps, box plots, and histograms reveal feature relationships, skewness, and outlier patterns.

Preprocessing

Missing value imputation, label encoding, one-hot encoding, and Min-Max / Standard scaling applied as appropriate.

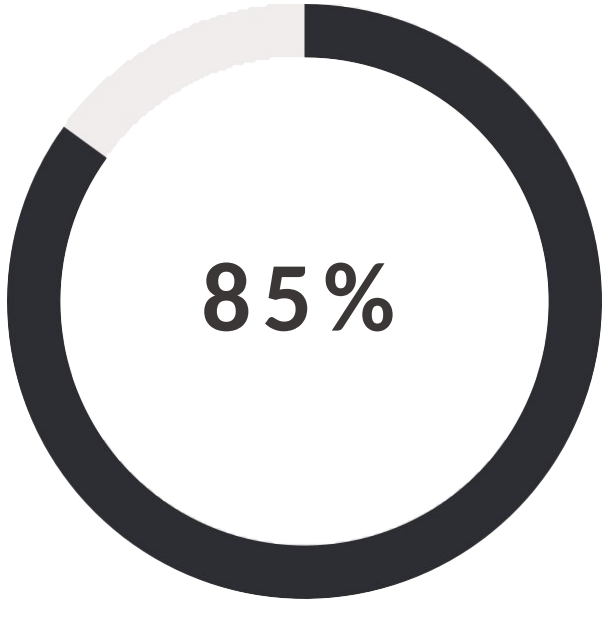
Forward Selection

Iterative greedy feature selection retains only those predictors that incrementally improve model R^2 on validation data.

Model Training

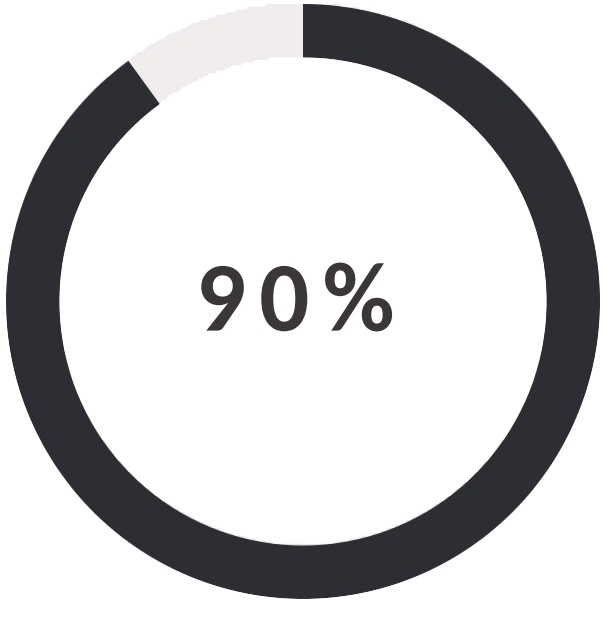
Linear, Ridge (L2), and Lasso (L1) regression models trained with cross-validation to optimise regularisation hyperparameters.

Expected Outcomes



Target R^2 Score

Models expected to explain ≥85% of salary variance on held-out test data.



Feature Reduction

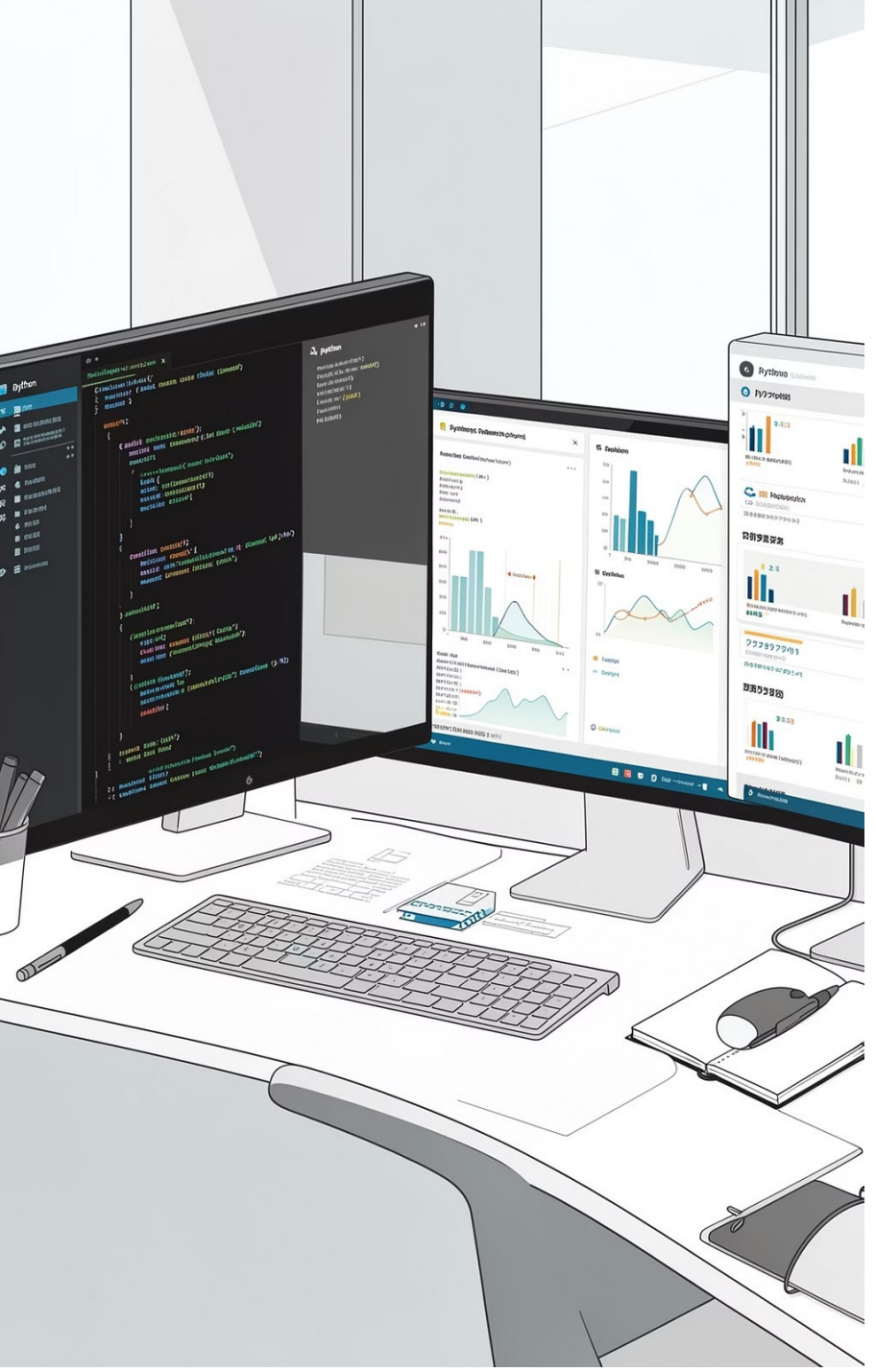
Forward Selection anticipated to retain ~9 of 20 features without meaningful accuracy loss.

Model Comparison Insight

Ridge regression is expected to outperform standard Linear Regression in the presence of multicollinearity, while Lasso will produce a sparser, more interpretable model — particularly valuable for HR practitioners.

HR Analytics Value

The final pipeline delivers a deployable salary estimation tool enabling fair, consistent, and transparent compensation decisions across departments and employment tiers.



Tools & Technologies

Python 3.x

Core programming language for all data processing, modelling, and visualisation scripts.

Pandas & NumPy

Data manipulation, synthetic generation, numerical operations, and array-based computations.

Scikit-learn

Regression model implementations, cross-validation, forward selection wrappers, and evaluation metrics.

Matplotlib & Seaborn

EDA visualisations including correlation heatmaps, residual plots, and feature importance charts.

Conclusion

This project demonstrates a **complete, end-to-end machine learning pipeline** — from synthetic data generation to model evaluation — addressing a meaningful real-world problem in HR analytics.

Academic Contribution

Validates regression-based approaches on controlled synthetic data with reproducible methodology.

Practical Impact

Provides HR teams with a transparent, auditable compensation prediction tool that reduces bias.

Extensibility

The pipeline architecture generalises to real organisational datasets and advanced ensemble methods.